

Previous versus Latest SapBERT Report

Three-level reference report compared with the current four-level study

This document compares your uploaded seven-page SapBERT_Full768_PCA64_Comparison-1.pdf with the eleven-page SapBERT_Final_Report.pdf. "Previous" means the earlier report you called ideal; it is a reference result, not a clinically established gold standard. No model was retrained for this document.

Main answer

Both studies test 1,999 records. Smaller confusion-matrix counts do not mean that the new model used fewer test cases. Class sizes, class definitions and which records enter the test set changed. Percentage-based scores really are lower for the new four-level task, but this is not a controlled measure of deterioration on the old task.

Metric (%)	Previous RF / PCA-64	Latest selected LR	Change (pp)
Accuracy	99.35	89.29	-10.06
Macro precision	99.04	89.67	-9.36
Macro recall	99.50	89.81	-9.69
Macro F1	99.26	89.73	-9.53

This headline compares different tasks and selected configurations. A negative difference means a lower reported score; it does not isolate the effect of adding one class. Same-family, fixed-configuration comparisons follow.

Question	Evidence-based answer
Are smaller matrix numbers bad?	Not by themselves. Compare correct / total within each row.
Are four levels better?	They meet the requested four-level specification and allow finer distinctions. Clinical usefulness is not established by the extra level alone.
Are the new scores as high?	No. The latest selected accuracy is 89.29%, versus 99.35% for the old RF task.
Has the latest refinement helped?	Versus the immediately previous four-level model: +0.30 pp accuracy and +0.30 pp macro F1. The development gain remains uncertain.

Reading guide: pp means percentage points; all precision, recall and F1 values are macro averages unless a class is named. Count tables and percentage matrices are deliberately distinguished. Differences are calculated before rounding, so subtracting displayed rounded values can differ by 0.01 pp.

Why the Confusion Counts Became Smaller

Property	Previous report	Latest report
Total source records	10,000	10,000
Development / test	8,001 / 1,999	8,001 / 1,999
Classes	1, 2, 3	0, 1, 2, 3
Test support by class	1: 677; 2: 943; 3: 379	0: 408; 1: 542; 2: 600; 3: 449
Matrix cells	$3 \times 3 = 9$	$4 \times 4 = 16$
Identical records in the two test sets	438	438

The largest old test class had 943 records. The largest new class has 600. A correct-count cell cannot exceed its row total. The old RF cell of 933 means $933 / 943 = 98.94\%$ recall. The latest Standard cell of 522 means $522 / 600 = 87.00\%$ recall. These are different class definitions, so the percentage difference is descriptive, not a paired class comparison.

Not every count decreased: the old RF class-3 diagonal is 379, whereas the latest Non-urgent diagonal is 411. Its row contains 449 records, so recall is 91.54% , below the old $379 / 379 = 100\%$. A larger count can coexist with a lower percentage.

The complete label transition (all 10,000 records)

Old label	New 0	New 1	New 2	New 3	Old total
1	2040	1349	0	0	3389
2	0	1365	2773	578	4716
3	0	0	227	1668	1895
New total	2040	2714	3000	2246	10000

Old class 1 splits into new 0 and 1. Old class 2 spreads across new 1, 2 and 3; old class 3 spreads across new 2 and 3. This is not a one-to-one renaming or simply an extra empty class. Subtracting cells by their numeric labels would be misleading.

The grouped stratified split was rebuilt for the new targets: 3,122 records change partition membership (1,561 in each direction). Both test sets still have 1,999 records, with 438 shared records. Earlier suggestions that a larger old test set explained these particular PDFs were not correct.

What Actually Explains the Score Gap?

A controlled diagnostic already separates target changes from split changes. It fits the old and new labels on the same current rows, split, SapBERT vectors, PCA-64, 22 patient features and fixed classifier settings. It is a retrospective diagnostic, not another model-selection round.

Classifier	Old-label accuracy	New-label accuracy	Change (pp)	Old / new F1
Logistic Regression	98.30	82.44	-15.86	98.11 / 83.03
HistGradientBoosting	99.20	82.09	-17.11	99.09 / 82.76
Random Forest	99.10	77.89	-21.21	98.96 / 78.35

The large gap remains when the split is held fixed. This supports a change in how predictable the supplied labels are from the available inputs. It does not prove that the new labels are wrong or establish a maximum achievable accuracy.

Observed change	Why a score can change	Judgment
Four target classes	More decision boundaries; distinctions previously inside one class can now count as errors.	Harder observed task; not automatically a worse design.
Different class proportions	Macro scores give each class equal weight, now one quarter rather than one third.	Different averaging problem; compare class recall too.
Changed test membership	Different cases can be easier or harder despite an identical test-set size.	Confounder in direct PDF comparisons.
Structured inputs matter	Old text-only accuracy is around 60%, but fusion is around 99%. Old labels are much more predictable with patient features.	Observed association, not proof of leakage or clinical correctness.
Concept compression	The existing audit flags lost or conflicting duration and other complaint details.	Plausible source of missing information; lexical flags need interpretation.
More PCA components / tuning	Latest model uses PCA-128, quadratic patient inputs and complaint details.	Measured improvement within the four-level task; no guarantee of 99%.

Renumbering identical classes alone would not change accuracy or macro F1. The transition table demonstrates that records were regrouped here. No investigation can assign an exact causal contribution to each changed prediction without the unavailable label-assignment rules.

Every Baseline Score: Text Only

Each row compares the same fixed classifier and input representation across the two reported tasks. A / P / R / F1 mean accuracy / macro precision / macro recall / macro F1. All entries are percentages; differences are new minus old, in pp. Recall comes from the saved prediction artifacts even where the old summary omitted it.

Condition	Metric	Previous 3-level	Current 4-level	Change (pp)
LOGREG / 768	Accuracy	60.08	44.17	-15.91
	Macro precision	64.40	44.89	-19.51
	Macro recall	58.52	44.87	-13.65
	Macro F1	60.63	43.95	-16.68
HGB / 768	Accuracy	60.73	41.52	-19.21
	Macro precision	67.31	43.72	-23.59
	Macro recall	57.70	40.95	-16.75
	Macro F1	60.33	41.70	-18.64
RF / 768	Accuracy	60.93	42.92	-18.01
	Macro precision	65.67	44.82	-20.84
	Macro recall	58.17	42.93	-15.24
	Macro F1	60.46	43.49	-16.97
LOGREG / 64	Accuracy	59.88	43.42	-16.46
	Macro precision	64.10	44.74	-19.36
	Macro recall	58.88	44.20	-14.69
	Macro F1	60.83	43.26	-17.57
HGB / 64	Accuracy	59.93	42.57	-17.36
	Macro precision	66.70	45.34	-21.36
	Macro recall	57.05	41.85	-15.19
	Macro F1	59.65	42.80	-16.85
RF / 64	Accuracy	59.53	40.62	-18.91
	Macro precision	63.00	41.88	-21.12
	Macro recall	57.46	40.93	-16.54
	Macro F1	59.31	41.08	-18.22

Why these values are lower: all these fixed-setting results face the new target definitions and a different grouped test membership. The controlled diagnostic confirms a substantial target-related gap, but cannot attribute a unique cause to each cell or decimal. Lower scores are worse agreement with their own reference labels, not proof that the encoder broke.

These are baseline conditions, not the latest tuned GUI model. Full 768-D and PCA-64 use the same frozen encoder. PCA reduces representation size; it does not create labels or improve results automatically.

Every Baseline Score: Text plus Patient Features

Each row compares the same fixed classifier and input representation across the two reported tasks. A / P / R / F1 mean accuracy / macro precision / macro recall / macro F1. All entries are percentages; differences are new minus old, in pp. Recall comes from the saved prediction artifacts even where the old summary omitted it.

Condition	Metric	Previous 3-level	Current 4-level	Change (pp)
LOGREG / 768	Accuracy	98.50	82.29	-16.21
	Macro precision	98.03	82.89	-15.15
	Macro recall	98.66	83.10	-15.56
	Macro F1	98.34	82.92	-15.41
HGB / 768	Accuracy	99.10	83.64	-15.46
	Macro precision	99.15	84.72	-14.43
	Macro recall	98.74	84.03	-14.72
	Macro F1	98.94	84.21	-14.74
RF / 768	Accuracy	99.35	77.44	-21.91
	Macro precision	99.04	77.87	-21.16
	Macro recall	99.50	78.24	-21.25
	Macro F1	99.26	77.93	-21.33
LOGREG / 64	Accuracy	98.50	82.44	-16.06
	Macro precision	98.03	83.01	-15.03
	Macro recall	98.66	83.22	-15.44
	Macro F1	98.34	83.03	-15.31
HGB / 64	Accuracy	99.15	82.09	-17.06
	Macro precision	99.24	83.28	-15.96
	Macro recall	98.78	82.54	-16.24
	Macro F1	99.01	82.76	-16.24
RF / 64	Accuracy	99.35	77.89	-21.46
	Macro precision	99.04	78.47	-20.57
	Macro recall	99.50	78.65	-20.85
	Macro F1	99.26	78.35	-20.92

Why these values are lower: all these fixed-setting results face the new target definitions and a different grouped test membership. The controlled diagnostic confirms a substantial target-related gap, but cannot attribute a unique cause to each cell or decimal. Lower scores are worse agreement with their own reference labels, not proof that the encoder broke.

These are baseline conditions, not the latest tuned GUI model. Full 768-D and PCA-64 use the same frozen encoder. PCA reduces representation size; it does not create labels or improve results automatically.

Tuned Families and the Latest Refinement

Old fused PCA-64 versus each latest tuned family. Settings and label task differ; these are descriptive changes, not an isolated classifier experiment.

Family / metric	Old PCA-64	Latest tuned	Change (pp)
Logistic Regression / Accuracy	98.50	89.29	-9.20
Logistic Regression / Macro precision	98.03	89.67	-8.36
Logistic Regression / Macro recall	98.66	89.81	-8.85
Logistic Regression / Macro F1	98.34	89.73	-8.60
HistGradientBoosting / Accuracy	99.15	88.19	-10.96
HistGradientBoosting / Macro precision	99.24	88.86	-10.38
HistGradientBoosting / Macro recall	98.78	88.58	-10.20
HistGradientBoosting / Macro F1	99.01	88.68	-10.32
Random Forest / Accuracy	99.35	82.29	-17.06
Random Forest / Macro precision	99.04	83.00	-16.03
Random Forest / Macro recall	99.50	82.76	-16.74
Random Forest / Macro F1	99.26	82.58	-16.68

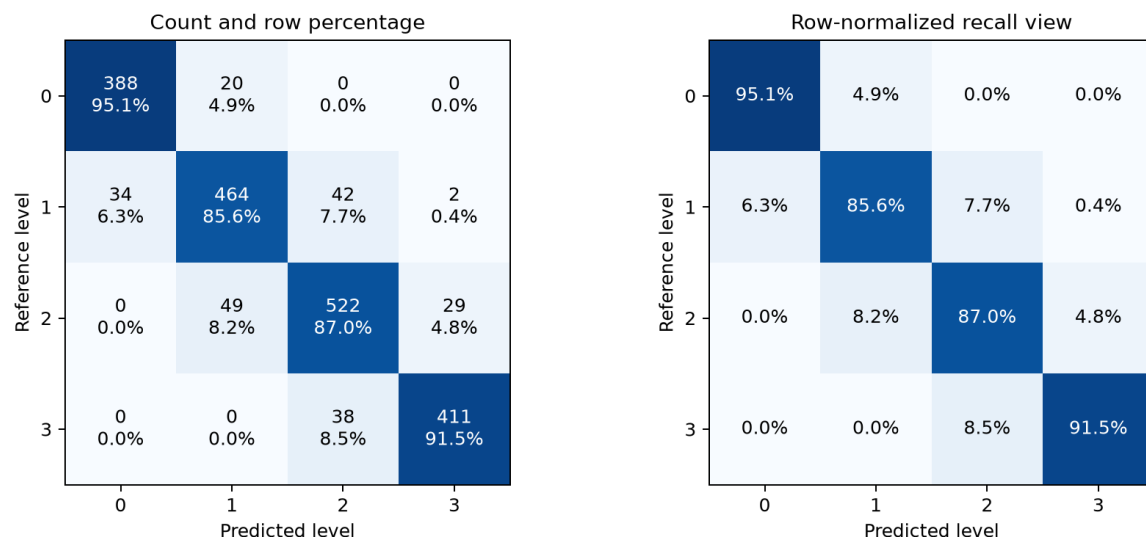
A fairer within-task comparison: immediately prior versus latest selected model

Metric	Initial ref.	Prior C=100	Latest C=10	Latest - prior
Accuracy	85.19	88.99	89.29	+0.30 pp
Macro precision	85.71	89.37	89.67	+0.31 pp
Macro recall	85.90	89.50	89.81	+0.31 pp
Macro F1	85.80	89.43	89.73	+0.30 pp
Emergency recall	92.16	93.87	95.10	+1.23 pp
Under-triage	7.60	4.65	4.65	+0.00 pp
Over-triage	7.20	6.35	6.05	-0.30 pp
Quadratic weighted kappa	0.9328	0.9497	0.9504	+0.0007
Mean absolute level error	0.1486	0.1106	0.1081	-0.0025

Initial reference means the original tuned PCA-64 LR C=10 model, not the fixed C=1 baseline. Higher accuracy, precision, recall, F1 and kappa are better; lower level error and under-/over-triage are better for these statistical measures. Under-triage is unchanged. The two models use the same rows and 197-feature design, with stronger LR regularization in the latest model. No label editing or SapBERT fine-tuning took place.

Development mean macro F1: 89.38% to 89.65%. The conditional paired gain interval is -0.08 to +0.64 pp and includes zero. The observed improvement is modest and not yet reliably established; repeated selection adds uncertainty beyond that interval.

Latest Four-Level Matrix: Counts and Meaning



Level	Support	Correct	Precision %	Recall %	F1 %
0 Emergency	408	388	91.94	95.10	93.49
1 Urgent	542	464	87.05	85.61	86.33
2 Standard	600	522	86.71	87.00	86.86
3 Non-urgent	449	411	92.99	91.54	92.26

Read every row

Emergency: 388 correct and 20 predicted Urgent. Urgent: 464 correct, 34 predicted Emergency, 42 Standard and 2 Non-urgent. Standard: 522 correct, 49 predicted Urgent and 29 Non-urgent. Non-urgent: 411 correct and 38 predicted Standard.

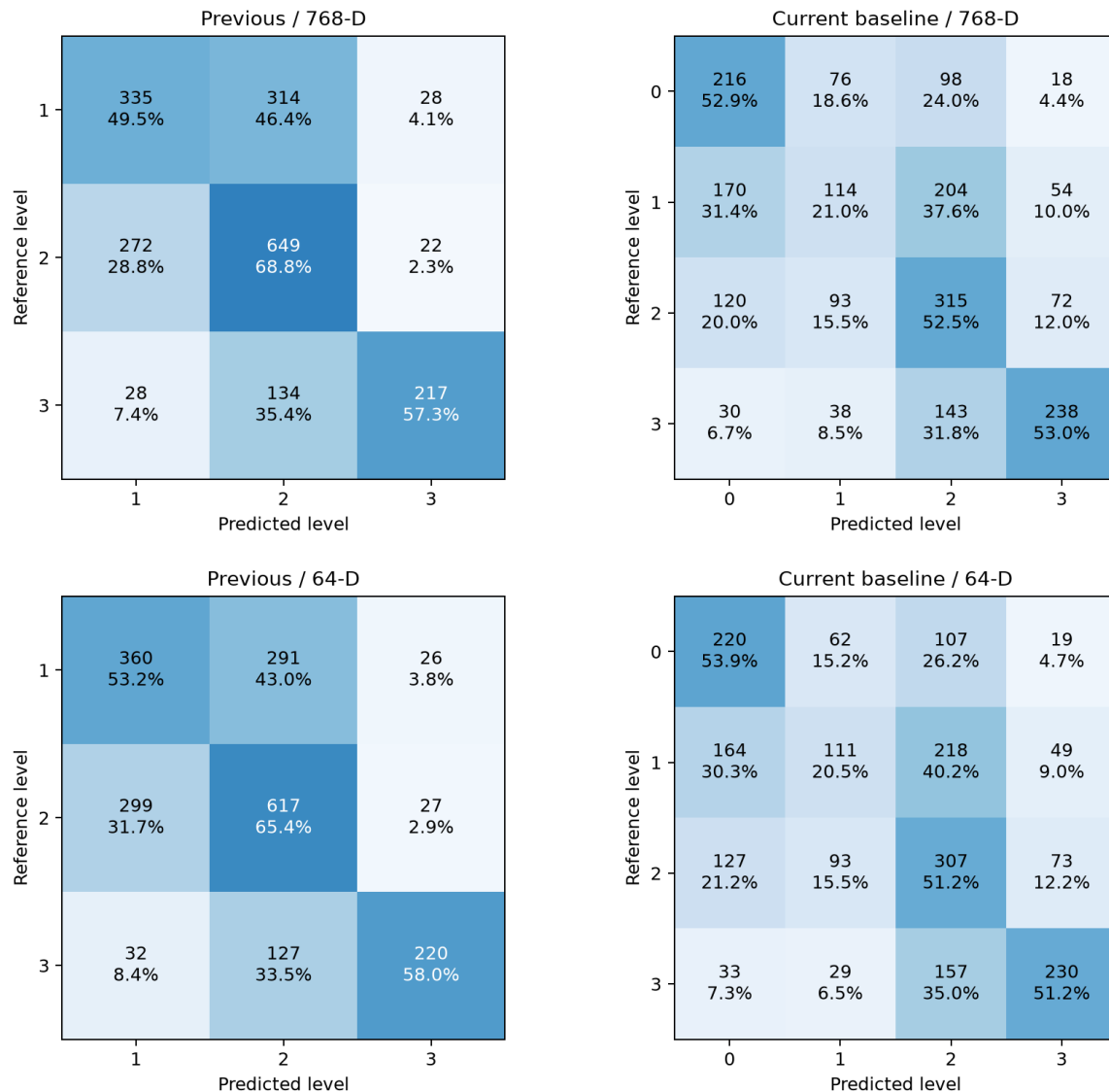
Total: 1785 correct / 1,999 = 89.29% accuracy; 214 errors. 212 errors are between neighbouring levels. Urgent and Standard have lower recall than the two outer classes. This identifies the main observed weakness; it does not justify changing reference labels.

Recall = diagonal / row total. Precision = diagonal / column total. F1 balances precision and recall. Macro averages weight each class equally. Adding a class changes these denominators and weights, so a raw count of 300 cannot be interpreted as "30% accuracy".

The old and new labels are not equivalent; class-by-class subtraction is therefore intentionally not presented. The following six pages include every old and new baseline matrix cell as a count and row percentage.

Logistic Regression: text only

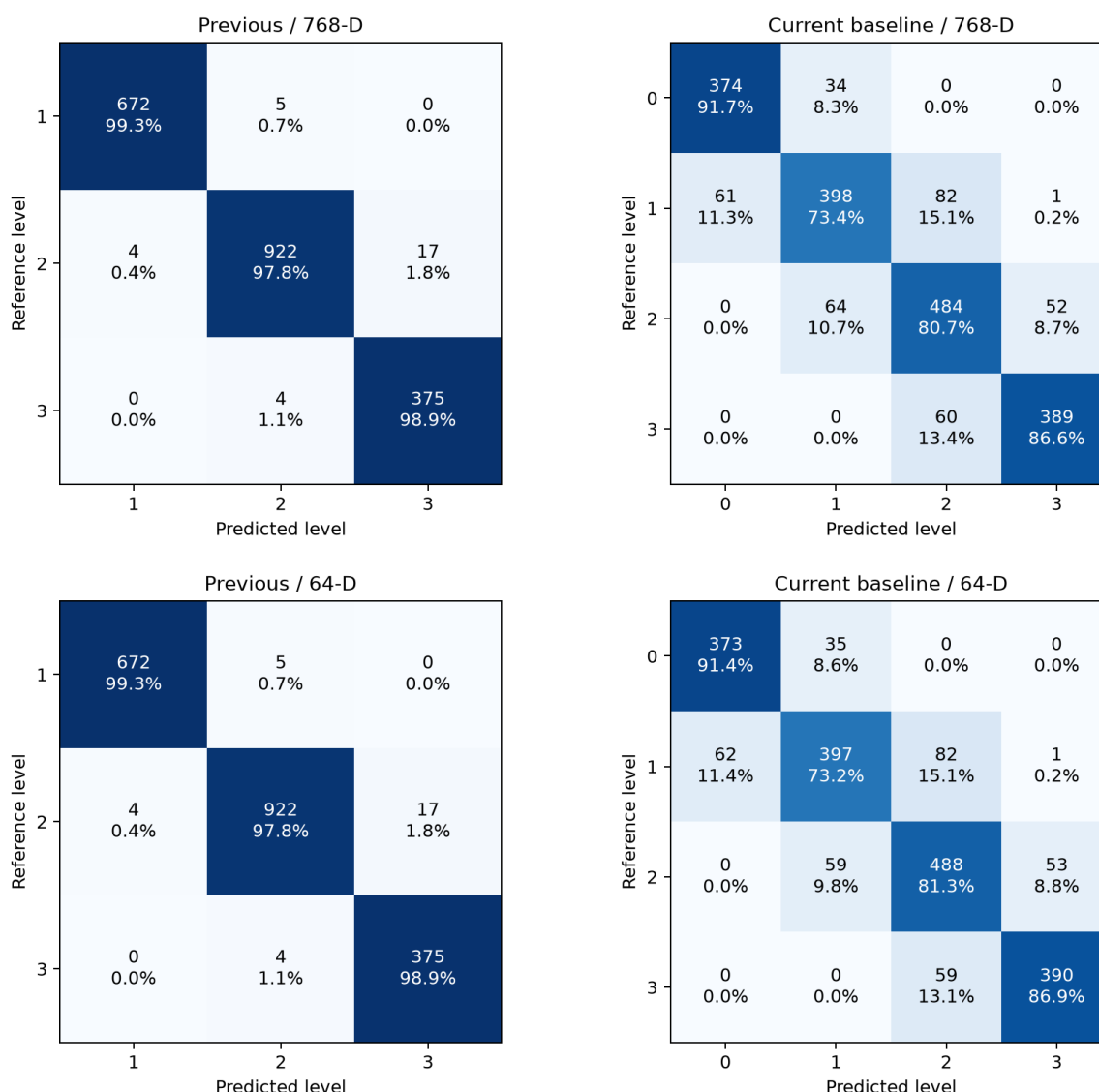
Previous three-level matrix on the left; current fixed four-level baseline on the right. Top: full 768-D. Bottom: PCA-64. Every matrix sums to 1,999. Each cell shows count and percentage of its reference row; color uses the same 0-100% scale.



Good direction: more of each row on its diagonal and less in off-diagonal cells. Smaller counts caused by smaller class support are neutral; lower diagonal percentages indicate lower recall on that task. Off-diagonal cells below the diagonal are over-triage; above it are under-triage within each ordered label system. Numeric labels must not be aligned across the two systems as if equivalent.

Logistic Regression: text + patient features

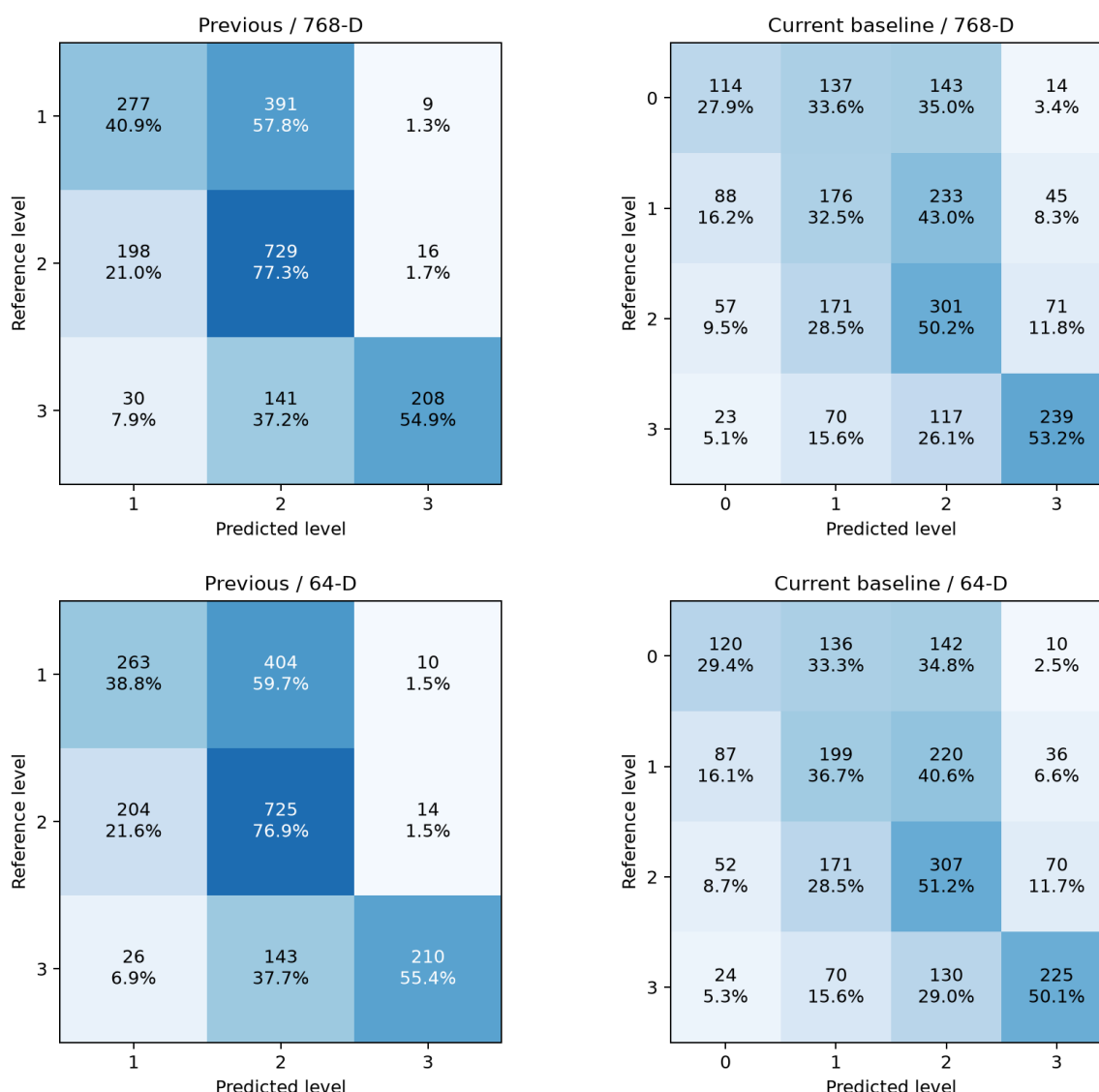
Previous three-level matrix on the left; current fixed four-level baseline on the right. Top: full 768-D. Bottom: PCA-64. Every matrix sums to 1,999. Each cell shows count and percentage of its reference row; color uses the same 0-100% scale.



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HistGradientBoosting: text only

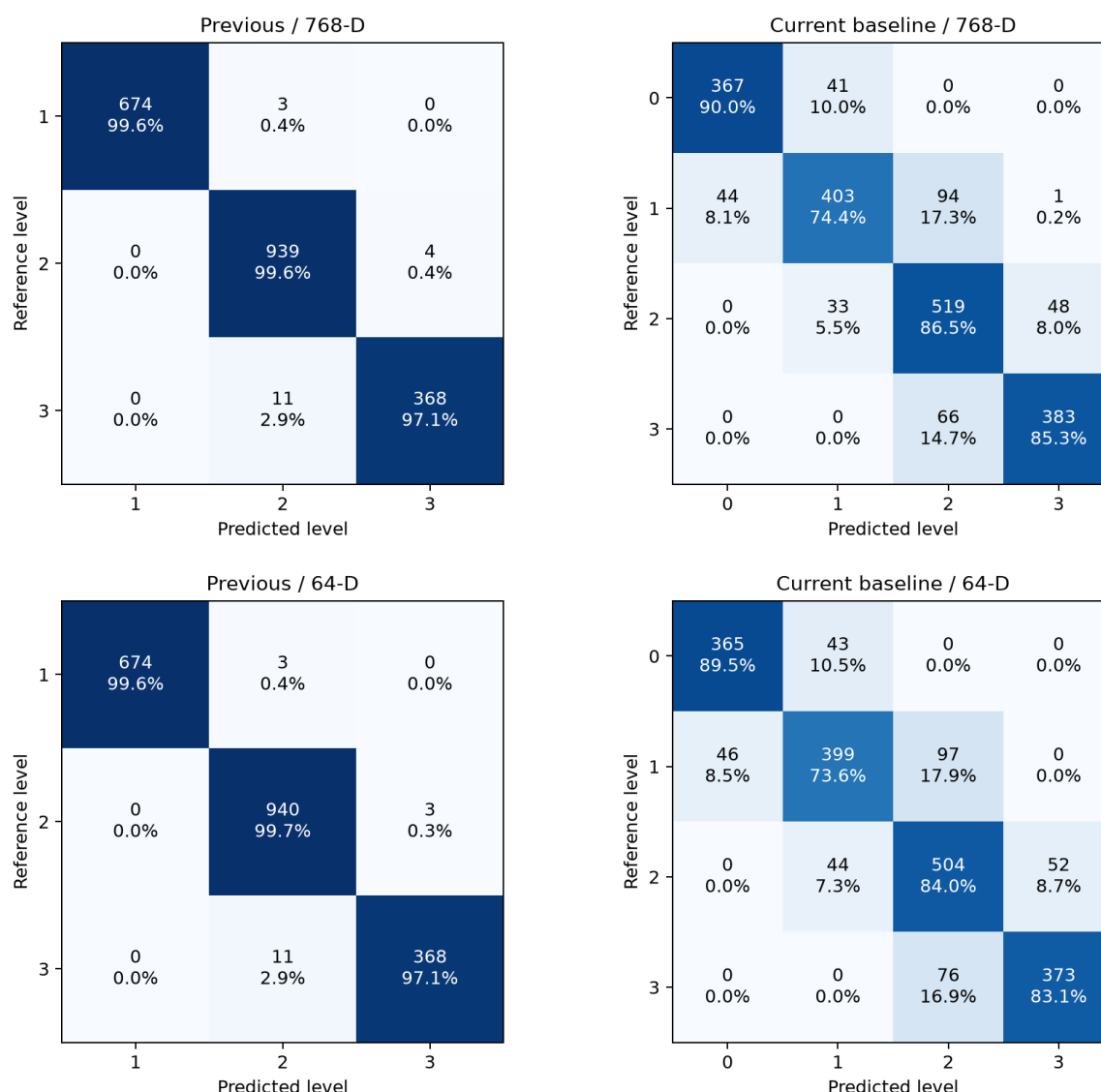
Previous three-level matrix on the left; current fixed four-level baseline on the right. Top: full 768-D. Bottom: PCA-64. Every matrix sums to 1,999. Each cell shows count and percentage of its reference row; color uses the same 0-100% scale.



Good direction: more of each row on its diagonal and less in off-diagonal cells. Smaller counts caused by smaller class support are neutral; lower diagonal percentages indicate lower recall on that task. Off-diagonal cells below the diagonal are over-triage; above it are under-triage within each ordered label system. Numeric labels must not be aligned across the two systems as if equivalent.

HistGradientBoosting: text + patient features

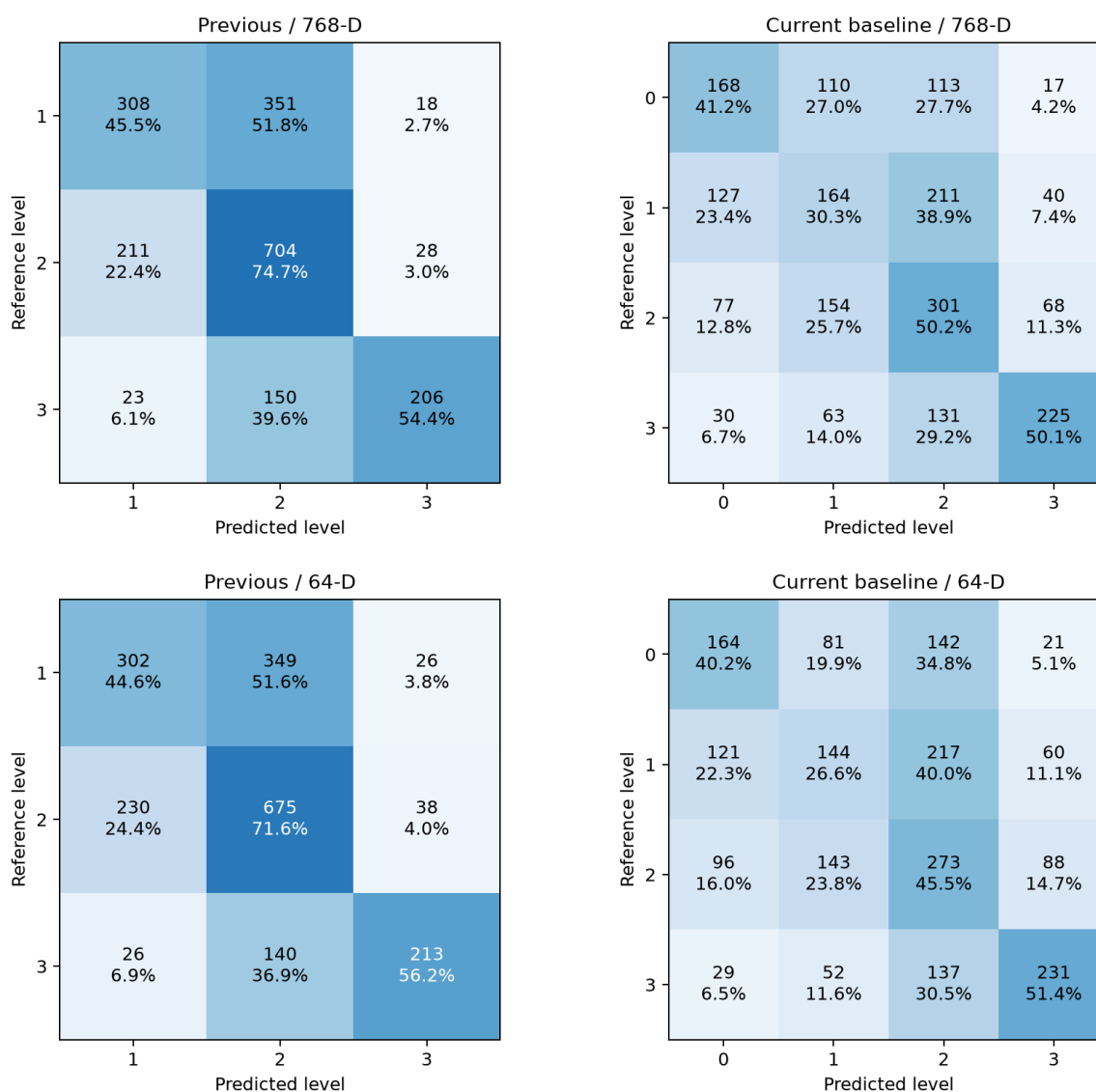
Previous three-level matrix on the left; current fixed four-level baseline on the right. Top: full 768-D. Bottom: PCA-64. Every matrix sums to 1,999. Each cell shows count and percentage of its reference row; color uses the same 0-100% scale.



Good direction: more of each row on its diagonal and less in off-diagonal cells. Smaller counts caused by smaller class support are neutral; lower diagonal percentages indicate lower recall on that task. Off-diagonal cells below the diagonal are over-triage; above it are under-triage within each ordered label system. Numeric labels must not be aligned across the two systems as if equivalent.

Random Forest: text only

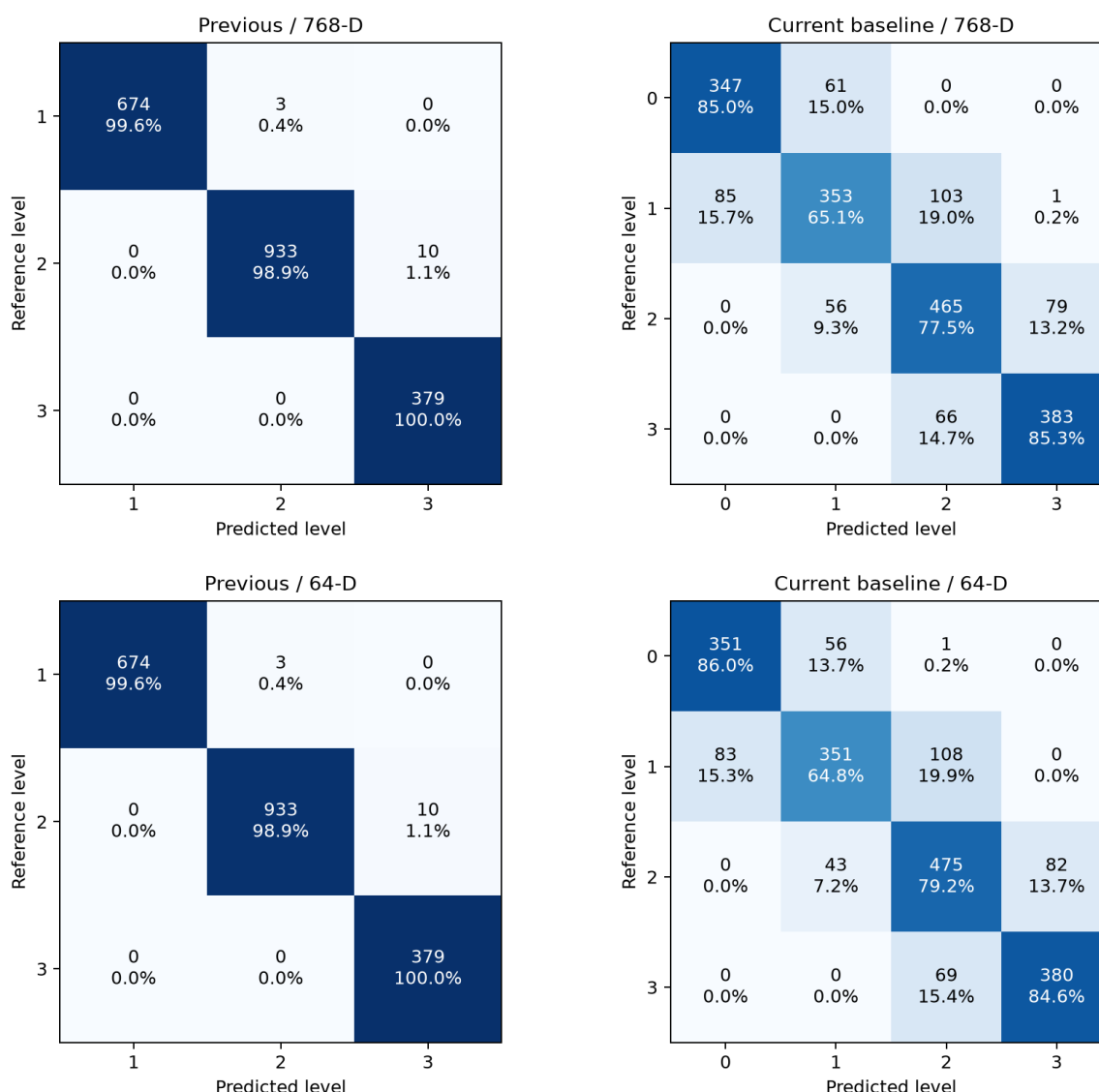
Previous three-level matrix on the left; current fixed four-level baseline on the right. Top: full 768-D. Bottom: PCA-64. Every matrix sums to 1,999. Each cell shows count and percentage of its reference row; color uses the same 0-100% scale.



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Random Forest: text + patient features

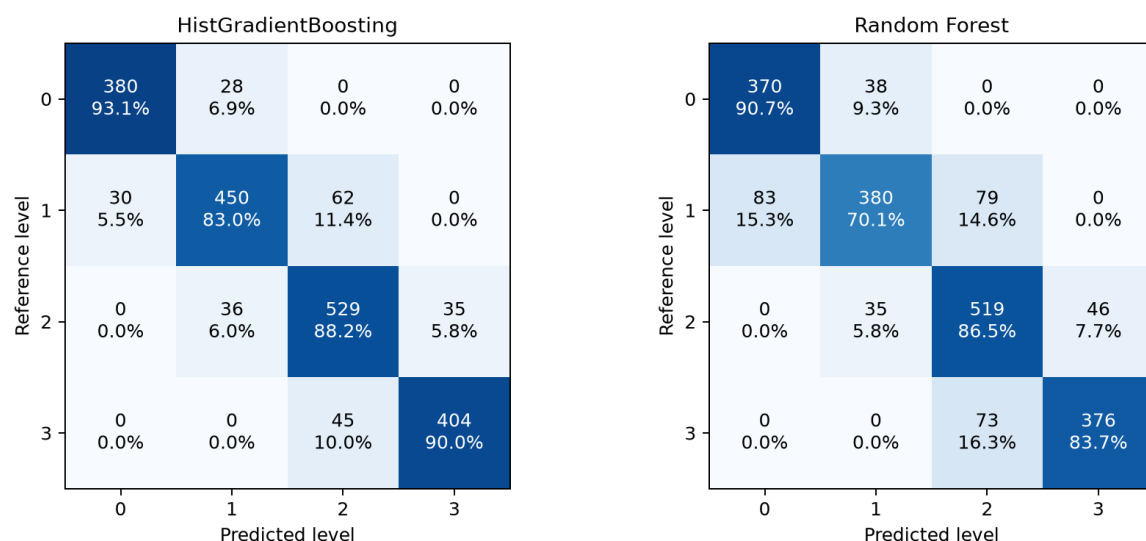
Previous three-level matrix on the left; current fixed four-level baseline on the right. Top: full 768-D. Bottom: PCA-64. Every matrix sums to 1,999. Each cell shows count and percentage of its reference row; color uses the same 0-100% scale.



Good direction: more of each row on its diagonal and less in off-diagonal cells. Smaller counts caused by smaller class support are neutral; lower diagonal percentages indicate lower recall on that task. Off-diagonal cells below the diagonal are over-triage; above it are under-triage within each ordered label system. Numeric labels must not be aligned across the two systems as if equivalent.

Latest Tuned HGB and Random Forest

These are the latest tuned four-level family results, rather than the fixed baselines on the preceding pages. Both use PCA-64 and original-complaint details. The deployed winner remains Logistic Regression; family comparisons do not override its selection rule.



Family / level	Support	Precision %	Recall %	F1 %
HGB / 0	408	92.68	93.14	92.91
HGB / 1	542	87.55	83.03	85.23
HGB / 2	600	83.18	88.17	85.60
HGB / 3	449	92.03	89.98	90.99
RF / 0	408	81.68	90.69	85.95
RF / 1	542	83.89	70.11	76.38
RF / 2	600	77.35	86.50	81.67
RF / 3	449	89.10	83.74	86.34

Each cell again shows its count and row percentage. The row totals are identical to those of the latest LR matrix, so comparisons among these three current family results share the same test records and targets. Their feature designs and tuned settings differ.

Latest HGB: 800 iterations, learning rate 0.05, 7 leaves, minimum leaf size 10, L2=5, balanced weights. Latest RF: 500 trees, minimum leaf size 2, feature fraction 0.3, balanced weights. The old fixed PDF used HGB 300 iterations / rate 0.1 and RF 300 trees / leaf size 3 / feature fraction 0.7. These setting changes are an additional reason not to attribute every old-to-new difference solely to the extra level.

Latest Development Search

All 68 settings and 340 full-size fits are retained. These five-fold development scores have no direct counterpart in the old fixed-comparison PDF. They must not be subtracted from old test scores. Values are percentages; SD is F1 fold standard deviation in pp.

Configuration	F1	SD	Accuracy	Emergency recall
lr_detail128_c10	89.65	0.26	89.28	94.06
lr_detail64_c100	89.56	0.82	89.19	93.93
logreg__complaint_details	89.38	0.50	88.99	93.63
lr_detail128_c30	89.31	0.38	88.91	93.93
lr_detail128_unbalanced	89.30	0.33	88.89	93.01
lr_detail128_c1000	89.08	0.52	88.66	93.26
lr_detail128_c300	89.00	0.57	88.58	93.44
lr_detail128_c1	88.68	0.63	88.29	94.42
lr_detail256_c100	88.65	0.36	88.21	93.01
hgb__complaint_details	88.07	0.95	87.65	92.89
logreg__concept_details	87.83	1.06	87.40	93.38
hgb_detail64_leaf7_l220	87.68	0.73	87.25	92.77
hgb_detail64_leaf15_l210	86.86	0.83	86.40	91.73
lr_quad128_c100	86.57	0.70	86.09	92.65
lr_pca128_c100_balanced0	86.57	0.59	86.05	91.60
lr_refine128_c300	86.50	0.75	86.00	92.52
lr_pca128_c1000_balanced1	86.48	0.69	85.98	92.28
lr_quad128_c300	86.48	0.68	85.99	92.34
lr_pca128_c100_balanced1	86.46	0.58	85.98	92.65
lr_quadratic64_c100_balanced1	86.41	0.80	85.93	92.83
lr_pca128_c10_balanced0	86.39	0.57	85.84	91.73
lr_refine128_c30	86.32	0.51	85.81	92.89
lr_quadratic64_c100_balanced0	86.31	0.51	85.81	91.05
lr_refine64_c30	86.26	0.61	85.75	93.50
lr_pca128_c1000_balanced0	86.24	0.47	85.71	91.05
lr_pca256_c10_balanced0	86.24	0.44	85.68	91.67
lr_quadratic64_c10_balanced0	86.23	0.66	85.71	91.24
lr_quadratic64_c10_balanced1	86.21	0.52	85.71	93.08
lr_pca64_c1000_balanced0	86.19	0.54	85.68	90.93
lr_pca64_c100_balanced1	86.19	0.46	85.69	93.01
lr_quad256_c100	86.18	0.92	85.66	91.54
lr_pca64_c100_balanced0	86.17	0.78	85.65	91.05
lr_pca128_c10_balanced1	86.16	0.43	85.63	93.07
lr_refine64_c300	86.06	0.52	85.55	92.83
lr_pca256_c100_balanced0	86.06	0.57	85.50	90.75

Highest mean macro F1 wins subject to the unchanged original emergency-recall threshold; accuracy breaks ties. The selected LR detail128 C=10 achieves 89.65% mean F1 and 94.06% emergency recall. More candidates can overfit development selection; these figures are not new independent validation.

Latest Development Search - Continued

All 68 settings and 340 full-size fits are retained. These five-fold development scores have no direct counterpart in the old fixed-comparison PDF. They must not be subtracted from old test scores. Values are percentages; SD is F1 fold standard deviation in pp.

Configuration	F1	SD	Accuracy	Emergency recall
lr_pca256_c10_balanced1	86.05	0.46	85.51	92.95
lr_pca64_c1000_balanced1	86.04	0.55	85.53	92.77
hgb_concept_details	86.02	1.03	85.54	91.48
lr_pca64_c10_balanced1	85.98	0.53	85.46	93.38
lr_pca256_c100_balanced1	85.91	0.70	85.36	91.97
lr_quad256_c300	85.90	1.07	85.35	91.24
lr_pca64_c10_balanced0	85.86	0.61	85.31	91.48
hgb_detail64_leaf31_l220	85.85	0.45	85.40	91.12
lr_pca256_c1000_balanced1	85.14	0.76	84.54	90.99
lr_pca256_c1000_balanced0	85.12	0.67	84.50	90.32
lr_pca128_whiten	85.06	0.46	84.51	92.34
lr_pca64_whiten	85.04	0.57	84.54	92.09
hgb_refine64_leaf7	84.76	0.80	84.26	90.38
hgb_fused_64_leaf7	84.34	0.65	83.81	91.36
hgb_fused_64_leaf15	84.24	0.76	83.74	90.81
hgb_fused_128_leaf15	83.75	0.59	83.23	90.44
lr_pca256_whiten	83.49	0.82	82.89	90.56
hgb_refine64_leaf31	83.45	0.63	82.94	90.14
hgb_fused_128_leaf7	83.41	0.98	82.83	91.61
lr_quadratic64_c1_balanced1	83.21	0.55	82.64	93.02
lr_quadratic64_c1_balanced0	83.20	0.73	82.56	90.26
rf_detail64_fraction0.3	81.96	0.63	81.58	91.79
rf_detail64_fraction0.7	81.55	0.88	81.10	88.73
rf_complaint_details	80.32	0.79	80.24	95.89
rf_concept_details	80.00	0.57	79.93	96.32
rf_refine64_leaf1	79.21	0.34	79.13	95.96
rf_refine64_leaf3	78.31	0.42	78.33	96.63
rf_fused_64	78.31	1.21	77.82	87.68
rf_refine128_leaf3	76.48	0.76	76.72	96.63
rf_refine128_leaf1	76.48	0.85	76.64	95.77
hgb_structured_0_leaf7	76.23	1.18	75.55	87.38
hgb_structured_0_leaf15	75.87	1.00	75.15	86.40
rf_structured_0	75.50	0.73	74.85	86.03

Highest mean macro F1 wins subject to the unchanged original emergency-recall threshold; accuracy breaks ties. The selected LR detail128 C=10 achieves 89.65% mean F1 and 94.06% emergency recall. More candidates can overfit development selection; these figures are not new independent validation.

Other Values: Features, Confidence and Audits

Quantity	Previous report	Latest selected model	Meaning
Embedding dimensions	768	768	Same frozen SapBERT encoder.
PCA dimensions	64	128	More retained components; not a performance percentage.
Patient / detail features	22 / 0	50 / 19	Quadratic numeric terms and original-complaint details.
Total fused inputs	790 full / 86 PCA	197	128 + 50 + 19. Larger is not automatically better.
PCA retained variance	91.31% at 64	95.51% at 128	Different PCA size/development partition; more variance is not clinical accuracy.

Probability diagnostic	Prior 4-level OOF	Latest OOF	Latest test
MCC	0.8522	0.8561	0.8563
Log loss	0.2648	0.2459	0.2419
Multiclass Brier score	0.1586	0.1523	0.1497
10-bin calibration gap	0.0331	0.0120	0.0094

MCC rises; log loss, Brier score and calibration gap fall: favorable descriptive changes in development predictions. These quantities were not reported in the old ideal PDF, so no old-versus-new probability comparison is invented. OOF means out-of-fold development predictions.

Audit values are not accuracy percentages

The pre-refinement model made 881 development errors: 878 adjacent-level errors, 354 Urgent/Standard errors, 220 errors with at least 90% confidence, and 345 errors shared by all three classifier families. These overlapping counts describe the prior model and must not be added together or presented as the latest test errors.

The input audit flags 567 duration disagreements, 142 durations not recovered in concepts, and 48 family-word omissions. Flags are not confirmed labelling errors. The learning audit has 90 fits: 60 nested-size learning fits and 30 detail comparisons. Its 30 full-size detail fits are already included in the combined 340; these totals must not be summed as independent experiments.

The latest PDF retains the learning audit unchanged. Its nine full-size conditions and graph values are reproduced on the next page so their older-stage meaning is explicit.

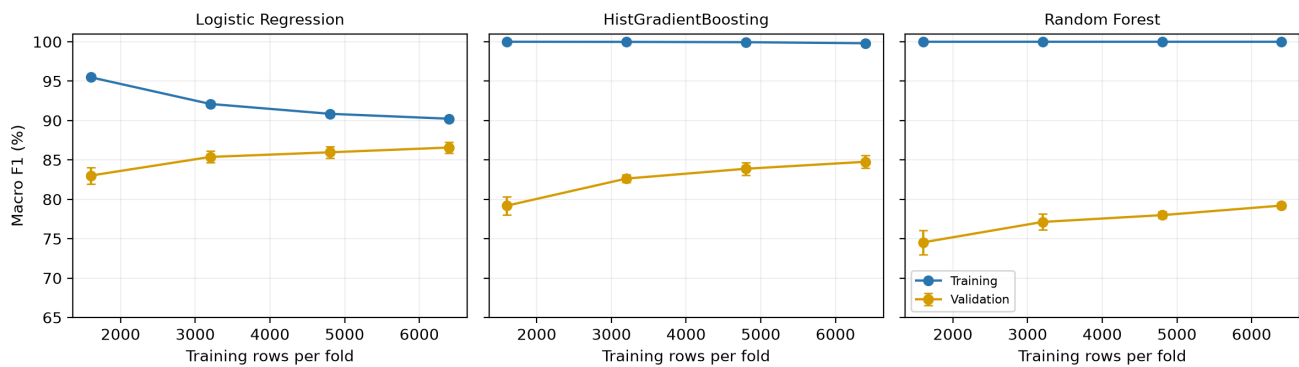
Latest development OOF	Pred. 0	Pred. 1	Pred. 2	Pred. 3
0	1535	97	0	0
1	141	1851	177	3
2	0	162	2090	148
3	0	0	130	1667

This matrix totals 8,001 development predictions, not 1,999 test predictions. Its larger cells do not imply higher performance; it evaluates a different sample through out-of-fold prediction.

Learning Audit and Literature: Unchanged Context

Audit condition	Accuracy %	F1 %	Emergency recall %
Logistic Regression / Prior features	86.09	86.57	92.65
Logistic Regression / Concept details	87.40	87.83	93.38
Logistic Regression / Original details	88.99	89.38	93.63
HistGradientBoosting / Prior features	84.26	84.76	90.38
HistGradientBoosting / Concept details	85.54	86.02	91.48
HistGradientBoosting / Original details	87.65	88.07	92.89
Random Forest / Prior features	79.13	79.21	95.96
Random Forest / Concept details	79.93	80.00	96.32
Random Forest / Original details	80.24	80.32	95.89

These are earlier development audit conditions, not the C=10 refinement. Nested subsets use 25%, 50%, 75% and 100% of each training fold; validation rows stay fixed. The learning curves suggest potential benefit from more distinct training information but do not predict a score at 20,000 rows.



Literature model	Accuracy %	Recall %	F1 %
TF-IDF + LR	75.00	98.80	54.40
TF-IDF + RF	75.10	96.40	58.20
BiLSTM	74.60	84.60	67.00
CNN	72.30	78.70	66.70

These four literature rows are unchanged between the reports: every delta is zero. They describe a different binary KTAS task and published metric definitions, not our four-level macro metrics. They cannot establish that either project model is clinically superior. Source carried forward from both PDFs: Seo et al. (2025), Scientific Reports 15, 16870, Table 2, DOI 10.1038/s41598-025-99874-0. This report does not newly audit that external study.

Conclusion: Which Changes Are Good or Bad?

Change	Assessment	Why
Lower confusion counts	Neutral on their own	Both tests have 1,999 records; per-class support changes. Use row percentages.
Lower old-to-new accuracy / F1	Lower agreement on a different task	The label transition changes what must be predicted. Controlled fits still show a large gap.
Four instead of three levels	Meets the requested specification	Finer output categories are possible, but correctness and clinical benefit require their own evidence.
Latest versus prior four-level scores	Modest observed improvement	Accuracy +0.30 pp; F1 +0.30 pp; emergency recall +1.23 pp. Development uncertainty includes no gain.
Urgent / Standard recall	Remaining weakness	Latest recall is 85.61% / 87.00%; neighbouring-level confusion dominates errors.
Separating experiment scopes	Necessary for interpretation	Do not describe old three-level 99% and current four-level results as the same experiment.
Claiming a definite cause for every error	Not supported	No assignment rules or independent review establish why each reference label was chosen.

Recommended wording for the article

"The previous three-level and current four-level experiments used the same number of test records but different target definitions and partly different test membership. A controlled comparison on a common split retained a substantial performance gap, indicating that the four-level targets were less predictable from the available inputs under the tested configurations. The final four-level model achieved 89.29% accuracy and 89.73% macro F1 in retrospective evaluation. These results are not a direct degradation estimate for the previous task or independent clinical validation."

Source and verification notes

Previous source: SapBERT_Full768_PCA64_Comparison-1.pdf, 7 pages. Latest source: SapBERT_Final_Report.pdf, 11 pages. All 24 baseline matrices and their headline metrics were recomputed from saved predictions; old displayed summary values were checked against PDF text. Current scores and the 68-setting table use verified round-five artifacts. The label-transition audit is historical evidence; this document runs no new training and edits no labels.

Internal evidence: output/results/triage_fixed_full_pca; reports/triage_four_level; reports/triage_four_level_round5; reports/label_transition_audit; reports/triage_error_investigation; reports/triage_learning_detail_audit. Unreported values are identified rather than guessed. No direct subtraction of non-equivalent class cells is made.